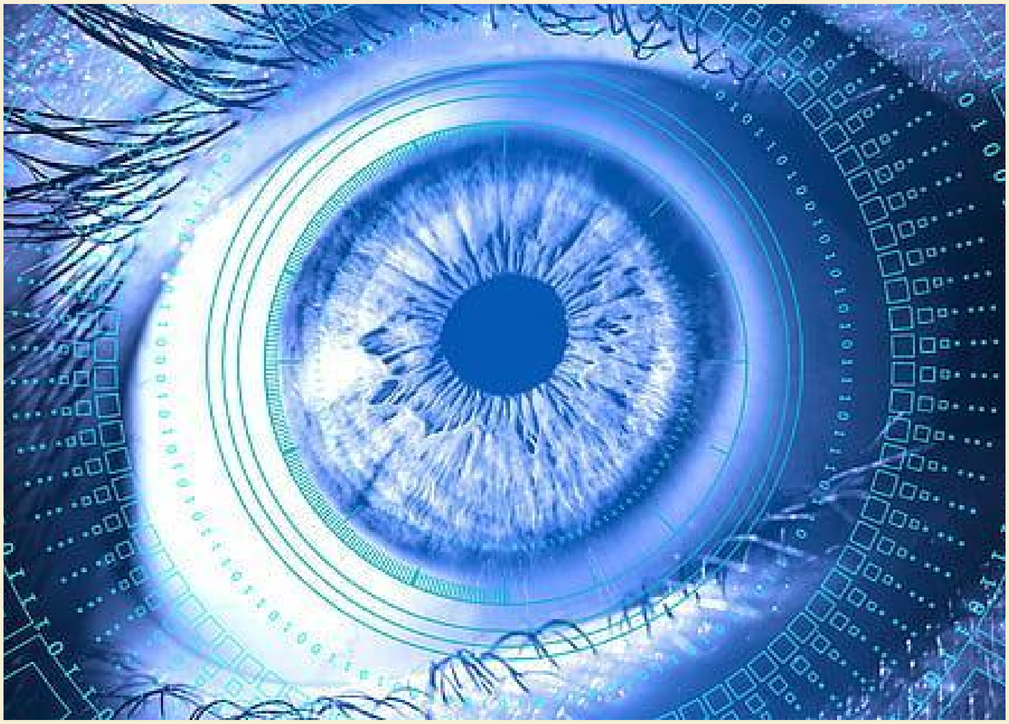
Proposal #25

Remote Eye-Tracking *Software Proposal*

CS Capstone Team 12



PFW CS Department -Sponsor

Prof. Jay Johns -Faculty Advisor

Dominic Minnich -Team Leader

Sulaiman Hussain, Logan Smith, Kyle Benich -Team Members

Abstract

This project proposes the development of a software tool utilizing standard computer or phone cameras to perform eye-tracking for remote user experience (UX) testing. Unlike conventional methods that require specialized equipment and physical labs, this solution aims to provide a cost-effective and accessible alternative for UX researchers. The primary objectives are to enhance remote UX testing capabilities, improve data accuracy by capturing real-time eye movements, and increase the accessibility of UX testing to a broader audience. The project will focus on developing a Flask server with key functionalities, such as user authentication, project creation, eye-tracking, and video recording. The anticipated outcome is a fully functional tool that delivers detailed insights into user behavior remotely for specific web pages or designs.

Problem Statement

Context and Background | Traditional UX testing relies heavily on physical labs and expensive specialized hardware, limiting its accessibility and scalability. With the rise of remote work and digital products, there is a growing demand for remote UX testing solutions that provide the same level of detail and accuracy without the associated costs.

Problem Definition | The challenge is to create a software tool that enables remote, accurate, and accessible UX testing using only standard computer or phone cameras, thereby eliminating the need for specialized equipment.

Relevance | This project is vital as it addresses the current limitations of UX testing, democratizing access to detailed user behavior insights. By making UX testing more accessible, the tool will allow companies, researchers, and students to enhance digital product usability, ultimately leading to improved user experiences.

Project Goals & Objectives

SMART Goals | Develop and deploy a functional eye-tracking tool by May.
Achieve a minimum of 90% data accuracy rate in eye-tracking within six months.

Long-Term Objectives | The project aims to establish the software as a leading tool for remote UX testing, continually improve its features, and expand its user base possibly past PFW.

Short-Term Objectives | The immediate focus is on building core functionalities, conducting initial testing, and gathering feedback to refine the product for a more functional release.

Scope

In-Scope | The project will encompass the development of a Flask server, implementation of eye-tracking using computer or phone cameras, creation of user roles, and project management features, along with real-time video recording with eye-tracking overlays.

Out of Scope | The project will not include the development of specialized eye-tracking hardware or any analysis unrelated to user experience.

Constraints | Constraints include limited budget and resources, dependency on open-source tools and technologies, and time limitations due to academic deadlines.

Assumptions | The project assumes that users will have access to a standard webcam or phone camera and a stable internet connection for data transfer.

Methodology

Approach | The project will utilize an Agile development methodology, which allows for iterative building and refinement of the software tool based on user feedback and testing.

Tools and Technologies | The development stack includes Flask, Python, Mediapipe (Google API), JavaScript, HTML, CSS, and SQLite for database management.

Data Sources | The project will leverage real-time eye-tracking data captured via webcams or phone cameras.

Development Process | The development process will involve several stages: planning, architecture design, core functionality development, eye-tracking integration, testing, and deployment.

Testing and Validation | Testing and validation will include unit tests, user acceptance testing (UAT), and scenario-based testing to ensure accuracy and reliability.

Timeline

Project Phases |
Planning: ~
Development: ~
Testing: ~
Deployment: ~

Milestones |
1. Completion of server setup.
2. Integration of eye-tracking functionality.
3. Initial user testing and feedback collection.
4.

Dependencies | The project relies on the availability of team members, access to hardware and software tools, and server stability.

Risk Analysis

Risk Identification | Potential risks include technical challenges with real-time data processing, privacy concerns regarding user data, and limited access to test users.

Risk Assessment | The technical challenges are of medium likelihood with high impact, while privacy concerns are of low likelihood but medium impact.

Mitigation Strategies | Regular code reviews, encrypted data handling, and clear consent protocols will be used to mitigate risks.

Stretch Goals

Mobile | Bring support for this web application to mobile in specific using the phones embedded camera for the project.

Heat Mapping | Extract heat maps from user eye tracking data on popular places the users are drawn to.

Domain Support | Support more than just the figma domain. This could be other established sites or custom sites.

Deliverables

Repository | Github repository containing version history, code sources, project timeline.

Documents | Documentation throughout our progress.